

Shashank Siddagangaiah

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Professional Summary

Principal AI & Data Engineer with a proven track record of managing high-performing engineering teams and delivering scalable, cloud-native AI solutions across the insurance, healthcare, and e-commerce sectors. Recognized for defining multi-year technical roadmaps, aligning complex data architectures with business strategy, and driving enterprise-wide digital transformations. Deeply specialized in developing Agentic AI frameworks, large-scale LLM integrations, and production-grade MLOps pipelines to power autonomous systems and predictive analytics.

Core Competencies & Technical Skills

- **Artificial Intelligence & Generative AI:** Large Language Models (LLMs), Agentic AI, Retrieval-Augmented Generation (RAG), NLP, LangChain, Hugging Face, AWS Bedrock, Google Vertex AI, Computer Vision, TensorFlow, PyTorch.
- **MLOps & Cloud Architecture:** Docker, Kubernetes, MLflow, Airflow, AWS (SageMaker, S3, Redshift, Lambda, Kinesis), Azure (Synapse, Data Factory), Google Cloud Platform (GCP), CI/CD.
- **Data Engineering & Big Data:** Databricks, Snowflake, Apache Spark (PySpark), Kafka, Medallion Architecture (Bronze-Silver-Gold), API-first Microservices, Hadoop, Delta Lake.
- **Programming & Databases:** Python (Pandas, scikit-learn), SQL (PostgreSQL, PL/SQL, T-SQL), Java, Scala, R, NoSQL (MongoDB, DynamoDB).
- **Leadership & Methodologies:** Agile, Scrum, SDLC, Technical Roadmapping, SLA Management, Data Governance as Code, Cross-functional Team Management.

Professional Experience

PEMCO Insurance / Bourntec Solutions | Chicago, IL | *Lead AI Engineer* | July 2025 – Present

- Directed the 1-to-3-year enterprise technical roadmap for AI capabilities and analytics infrastructure, strategically aligning engineering deliverables with core business retention goals.
- Architected and deployed a cutting-edge Agentic AI framework to autonomously analyze Customer Lifetime Value (CLV), integrating complex underwriting, pricing, and claims data.
- Engineered intelligent, self-learning AI data agents and API-first microservices that dynamically retrieved and linked multi-system datasets, reducing manual validation and improving data linkage accuracy by 25%.
- Designed and orchestrated end-to-end Medallion (Bronze-Silver-Gold) data architectures within Databricks and Microsoft Fabric, guaranteeing 99.9% data availability and strict SLA adherence for mission-critical ML workloads.
- Built automated ETL/ELT pipelines using Fabric Data Factory and Synapse Analytics, leveraging parallelized orchestration to cut pipeline latency by 30%.
- Embedded comprehensive policy-as-code controls, schema enforcement, and lineage tracking directly into AI pipelines to ensure robust data quality and governance.

Kemper Insurance / Bourntec Solutions | Chicago, IL | *Lead Data Engineer* | Aug 2023 – July 2025

- Managed an Agile engineering team of 8 direct reports, driving quarterly performance reviews, technical mentorship, and execution of the cloud data roadmap.
- Architected highly scalable enterprise data warehouse solutions in Snowflake and constructed real-time ingestion pipelines utilizing Apache Kafka, Spark Streaming, and AWS Kinesis.
- Spearheaded the legacy modernization of the enterprise data ecosystem, migrating 70+ OBIEE/MicroStrategy workloads to Informatica Cloud and Power BI, slashing licensing costs by 40% while boosting refresh speeds by 50%.
- Developed scalable, object-oriented data applications in Java, Scala, and Python, standardizing transformation methodologies and accelerating new pipeline development by 25%.
- Engineered resilient pipeline orchestration systems utilizing Apache Airflow, which reduced system downtime by 30% and enabled strict operational SLA tracking.
- Enforced stringent enterprise privacy requirements (HIPAA, GDPR) by instituting robust PII masking strategies and role-based access controls (RBAC) across Snowflake environments.

Select Rehabilitation | Chicago, IL *Data Engineer / Data Analyst* | May 2022 – Aug 2023

- Designed and productionized scalable ETL/ELT workflows using Databricks, Spark, AWS Glue, and Azure Data Factory, accelerating healthcare data processing throughput by 30%.
- Containerized data-processing applications utilizing Docker and Kubernetes to facilitate microservice orchestration and seamless deployment automation.
- Transformed unstructured MongoDB healthcare datasets into optimized relational star-schemas to support real-time operational reporting.
- Led the migration of 100+ legacy IBM Cognos BI reports into a proprietary Python-based platform, cutting operational costs by 20%.

Amazon | Bangalore, India *Senior Pricing Data Analyst / Business Analyst* | May 2016 – Dec 2020

- Partnered with global Data Scientists and Software Engineers to design backend architectures supporting the international expansion of Amazon Fresh across the US, UK, Japan, and Germany.
- Developed advanced machine learning models (Random Forest, XGBoost) in Python and R for dynamic pricing optimization and rigorous demand forecasting.
- Built automated Python-based analytics pipelines and backend data services, resulting in a 30% increase in data refresh efficiency for near-real-time business insights.
- Spearheaded a massive unstructured data integration initiative via APIs during the COVID-19 pandemic, resolving complex refund tracking issues and reducing customer service contacts by over 90% (from 3,000 to 200 weekly).
- Engineered massive datasets utilizing advanced SQL and AWS Redshift to support the cross-functional engineering requirements of the Pricing Analytics and Amazon Pay teams.

Education & Certifications

Master of Science in Business Analytics (GPA: 3.9) | *University of Illinois Chicago*, Chicago, IL | Jan 2021 – May 2022

Certifications: * AWS Certified Data Engineer – Associate

- Tableau Desktop Specialist